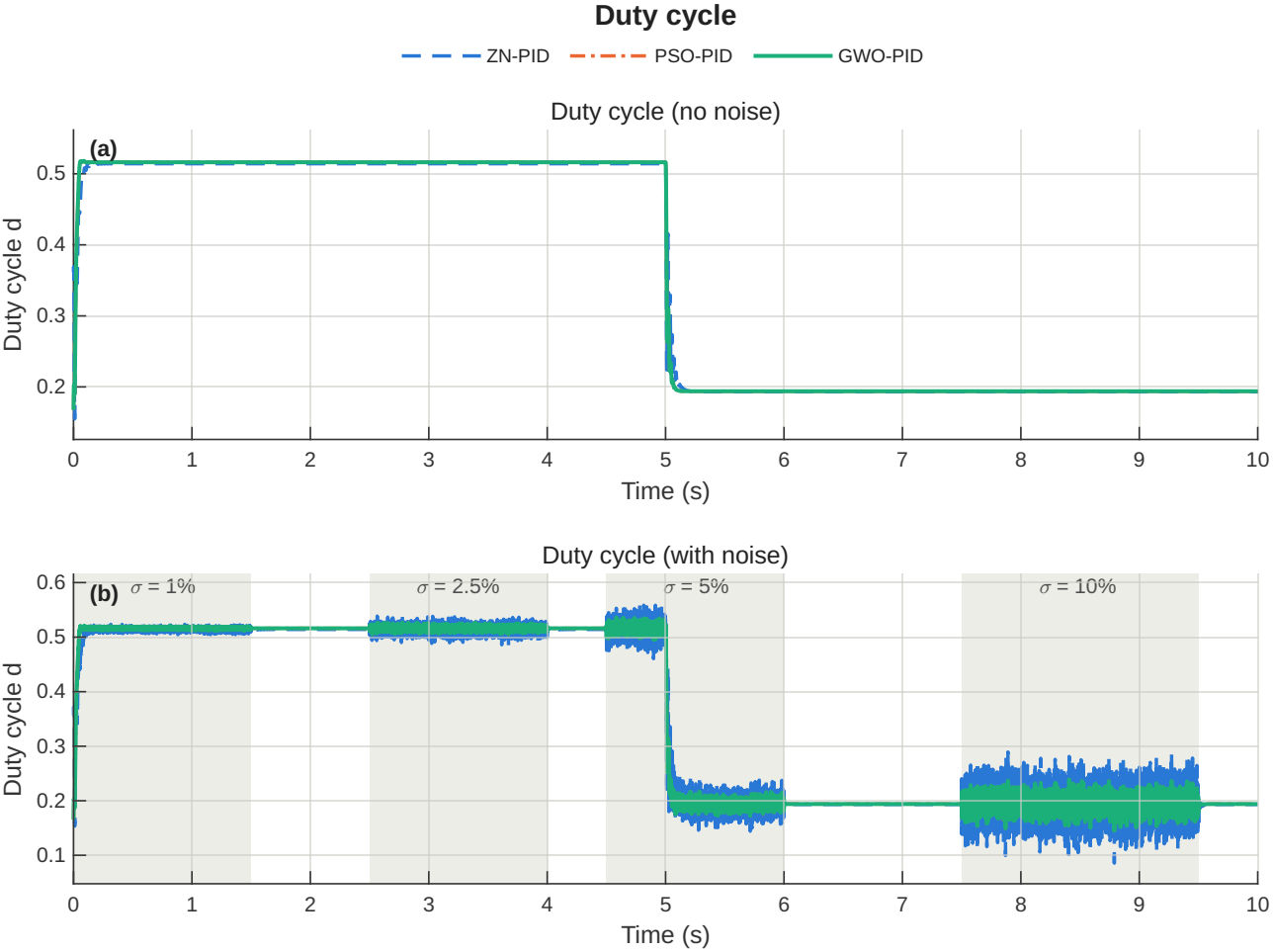

Duty cycle
ZN-PID
PSO-PID
GWO-PID
Duty cycle (no noise)
(a)
Duty cycle d
0.5
0.4
0.3
0.2
0
1
2
3
4
5
6
7
8
9
10
Time (s)
Duty cycle (with noise)
(b)
σ = 1%
σ = 2.5%
σ = 5%
σ = 10%
Duty cycle d
0.6
0.5
0.4
0.3
0.2
0.1
0
1
2
3
4
5
6
7
8
9
10
Time (s)